

Water Tank Water Level Indicator

An innovative solution for efficient water management, preventing wastage and conserving energy in various applications.

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INTRODUCTION : Water wastage is a serious problem Overflow of water tanks causes loss of water and electricity.

Significant Loss

Water wastage is a critical issue globally, contributing to resource depletion and increased utility costs.

Overflow Consequences

Traditional water tank systems often lead to overflow, resulting in substantial water loss and unnecessary electricity consumption by pumps.

Our Solution

This project offers a practical and immediate solution by clearly indicating water levels and alerting users to prevent overflow.

Project Objectives: Precision and Simplicity

01

Accurate Level Detection

To precisely sense and monitor the water level within a storage tank.

02

Visual Water Status

To provide clear, visual feedback of water levels using a series of LED indicators.

03

Audible Overflow Alert

To incorporate an audible buzzer that signals when the water tank has reached its maximum capacity.

04

Cost-Effective Design

To engineer a solution that is both affordable and straightforward to implement for widespread adoption.

Versatile Applications: Where It Makes a Difference



Domestic Use

Ideal for residential water tanks, ensuring households manage water efficiently and avoid wastage.



Multi-Dwelling Units

Essential for hostels, apartment complexes, and other large residential buildings to streamline water distribution.



Industrial Systems

Applicable in factories and industrial settings for monitoring large-scale water storage, improving operational efficiency.



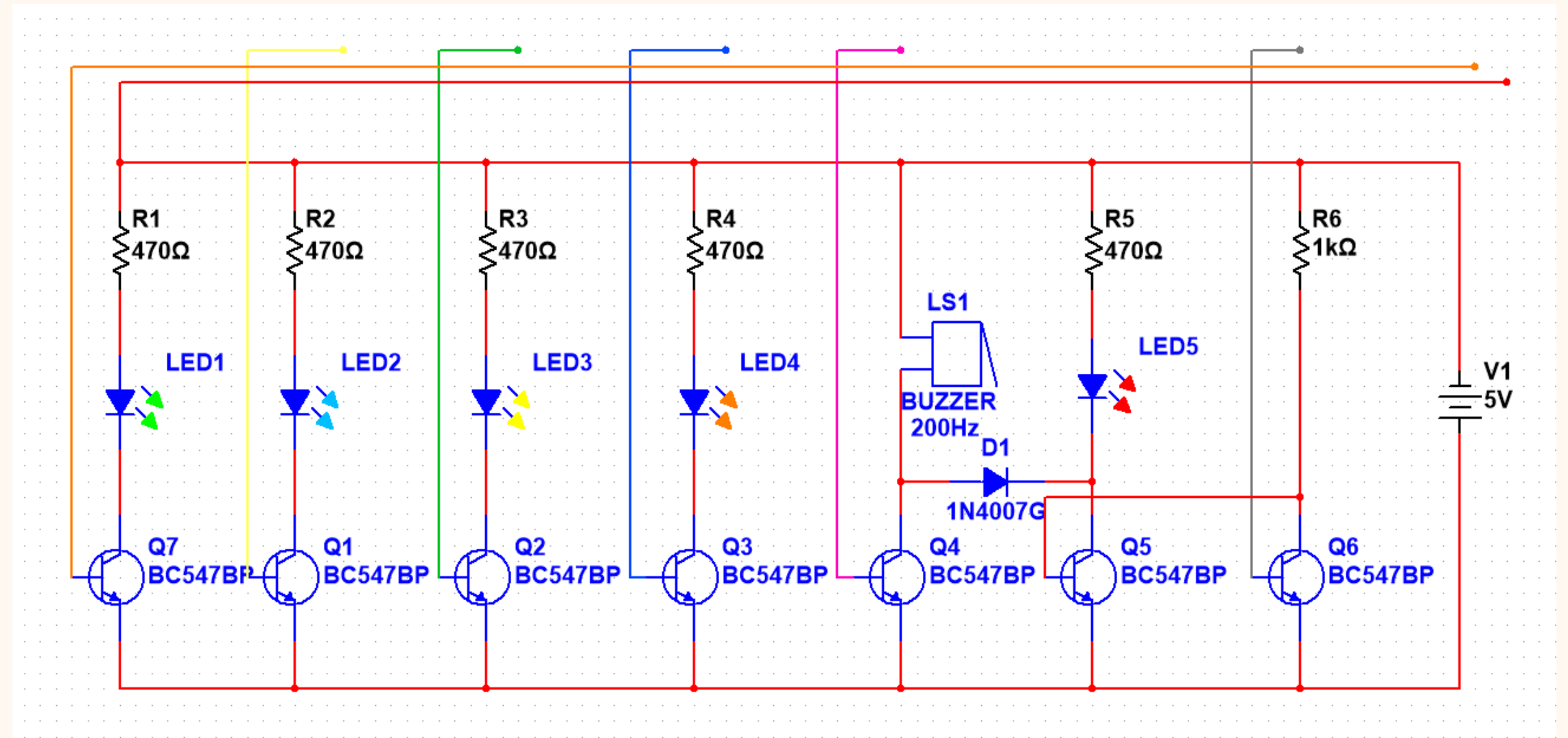
Agricultural Sector

Beneficial for irrigation systems, helping farmers optimise water usage and reduce manual checks.

Components:

- BC547 NPN Transistors (**7 pieces**)
- 470 Ω Resistors (**5 pieces**)
- 1k Ω Resistor (**1 piece**)
- 1N4007 Diode (**1 piece**)
- 5V Buzzer (**1 piece**)
- 5mm LEDs: Red, Orange, Yellow, Green, Blue (**1 of each**)
- Breadboard
- Jumper Wires

Circuit Diagram:



Component Deep Dive: Essential Roles



BC547 NPN Transistor

Functions as a switching device. A small current at its base allows a larger current to flow from collector to emitter, effectively driving the LEDs and buzzer.



Resistors (470 Ω , 1k Ω)

Crucial for current limitation. 470 Ω resistors protect LEDs, while the 1k Ω resistor regulates the base current of the transistor, safeguarding components.



1N4007 Diode

Provides vital circuit protection by preventing reverse current flow and mitigating back electromotive force (EMF) generated by the buzzer, enhancing reliability.

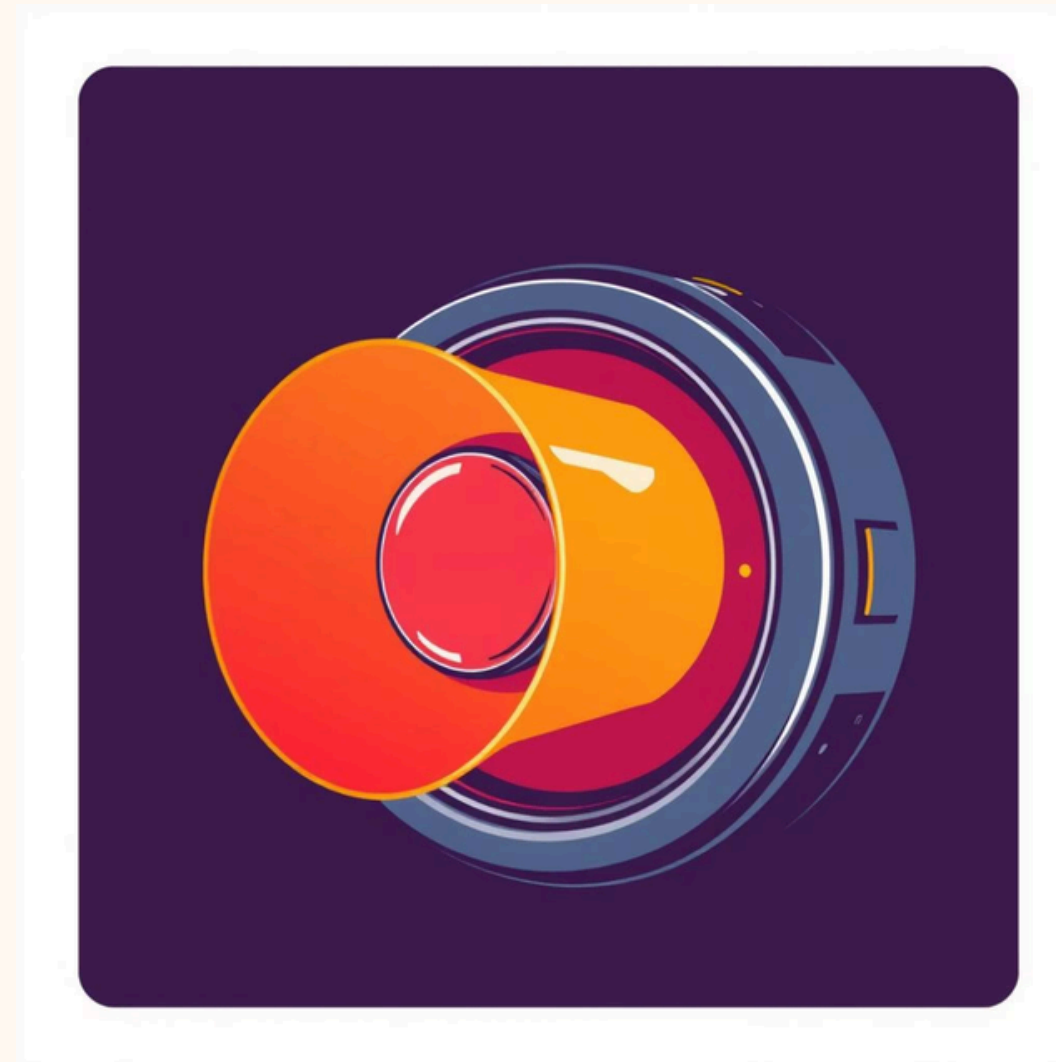
Visual & Audible Alerts: Keeping You Informed



LED Indicators

- **Low Level:** Red LED
- **25% Full:** Orange LED
- **50% Full:** Yellow LED
- **75% Full:** Blue LED
- **100% Full:** Green LED

These provide an intuitive visual representation of the water level at a glance, ensuring clear status updates.



5V Buzzer

- Activates automatically when the tank is completely full.
- Emits a clear audible alert to notify the user immediately.
- Effectively prevents costly and wasteful water overflow.

How It Works: Principle of Operation



Water Conductivity

The system leverages the natural conductivity of water. When water makes contact with the sensing probes, it completes a circuit.



Current Activation

The flow of current through the water activates the base of the transistor, causing it to switch to an ON state.



LED Illumination

Once the transistor is active, the corresponding LED connected to its collector terminal illuminates, indicating the water level.



Full Tank Alert

Upon reaching the maximum water level, the final transistor activates the buzzer, providing an unmistakable audible warning.

Step-by-Step: The Working Process

1

Initial Contact

Water gradually rises and touches the lowest sensing probe, initiating the circuit response.

2

Red LED On

The red LED lights up, signalling the first detected water level.

3

Progressive Illumination

As water continues to fill, subsequent sensing probes are triggered, illuminating higher LEDs sequentially.

4

Buzzer Activation

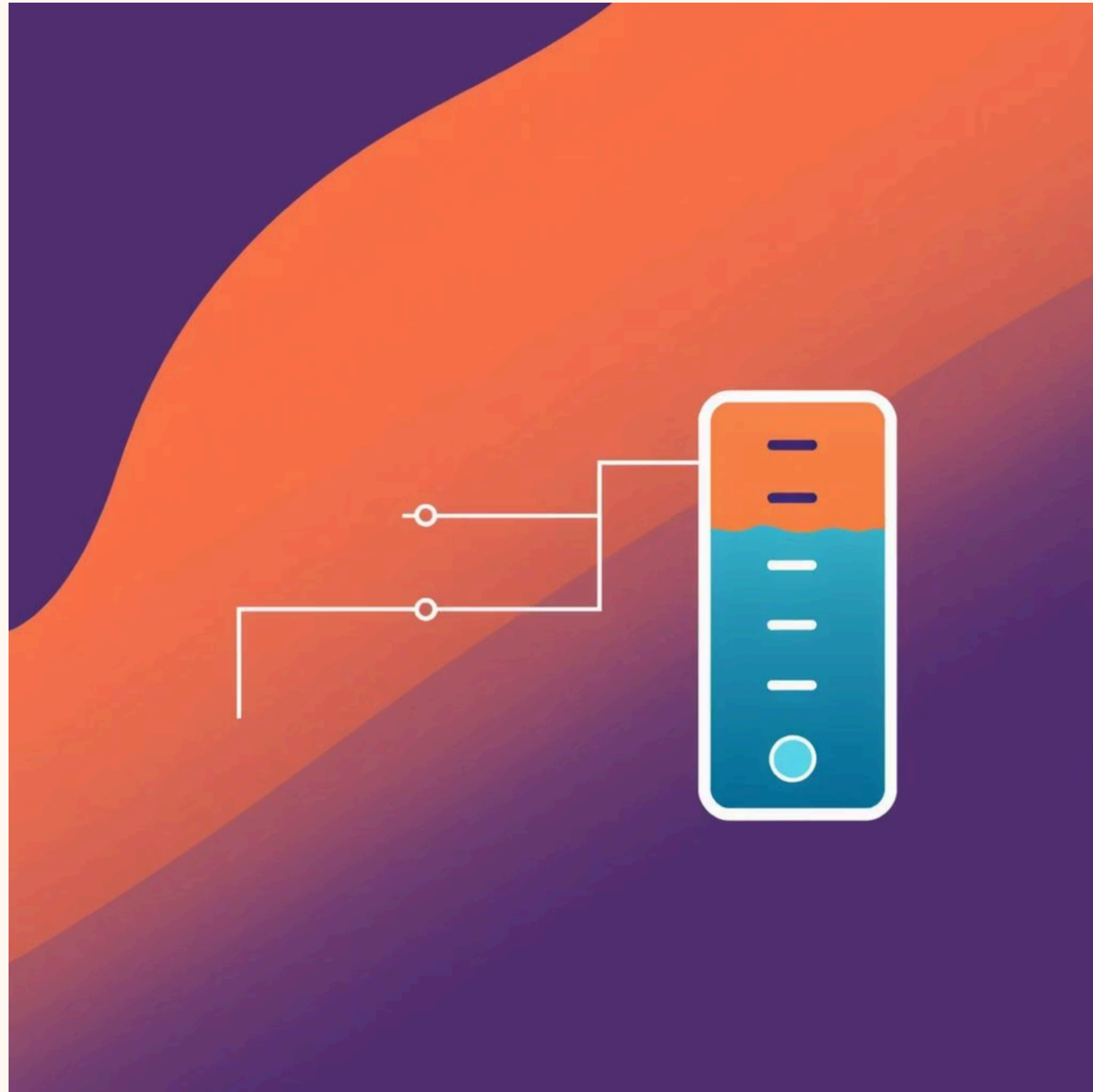
When the water reaches the highest probe, the buzzer activates, indicating the tank is full.

5

User Intervention

The user is alerted by the buzzer and can manually switch off the water pump, preventing overflow.

Conclusion & Overview



- **Individual Transistors:** Each water level sensor is connected to its own dedicated transistor for independent detection.
- **LED Connections:** The indicator LEDs are strategically connected to the collector terminals of their respective transistors.
- **Common Ground:** A unified common ground ensures stable and reliable operation across the entire circuit.

This project is very useful in real-life applications. Water level monitoring is achieved using a simple method & helps to prevent overflow and water wastage.